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Local Control on Fracture Patterns in Salt Dome Area

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Salt halokinesis and the widespread development of salt domes in the southwest of the Arabian Gulf have significantly influenced the regional stress regimes, resulting in diverse and complex fracture patterns, particularly within the tight carbonate reservoirs of the Upper Jurassic (Noufal and Shebl, 2019). These fractures play a critical role in reservoir performance, yet their characterization remains a challenge due to the multi-phase tectonic overprints and the structural complexity introduced by salt-related deformation (Fig. 1).

This study aims to investigate the fracture system developed above salt domes under Jurassic rifting, Late Cretaceous transtension, and Tertiary compression. By integrating seismic attributes, borehole imaging logs, and structural evolution analysis, we construct a conceptual fracture model to interpret local fracture patterns and validate them against dynamic data. The results demonstrate that fracture intensity and geometry vary with structural position around the salt dome—major fractures concentrate on the flanks in oval geometries, while crests remain less fractured. Additionally, NW-SE en-echelon fractures from Late Cretaceous transtension dominate areas away from the dome, and their overlap with salt-induced fractures creates highly complex systems in transitional zones. This work provides a practical workflow and geological insights for understanding and predicting fracture development in salt-influenced tight carbonate reservoirs.

Introduction

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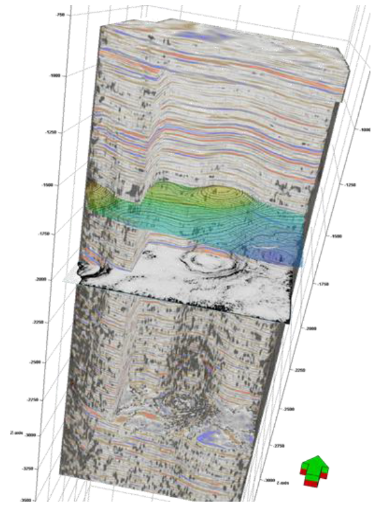

Figure 1—Salt halokinesis and salt domes in the Arabian Gulf.

This study aims to investigate the fracture system developed above salt domes under Jurassic rifting, Late Cretaceous transtension, and Tertiary compression. By integrating seismic attributes, borehole imaging logs, and structural evolution analysis, we construct a conceptual fracture model to interpret local fracture patterns and validate them against dynamic data. The results demonstrate that fracture intensity and geometry vary with structural position around the salt dome—major fractures concentrate on the flanks in oval geometries, while crests remain less fractured. Additionally, NW-SE en-echelon fractures from Late Cretaceous transtension dominate areas away from the dome, and their overlap with salt-induced fractures creates highly complex systems in transitional zones. This work provides a practical workflow and geological insights for understanding and predicting fracture development in salt-influenced tight carbonate reservoirs.

Structural evolution analysis

Three major tectonic events have shaped the structural framework of the Arabian Plate: Jurassic rifting, Late Cretaceous transtension, and Tertiary compression (Richard et al., 2014; Noufal et al., 2016). The Jurassic rifting (~150 Ma) initiated a NW-SE trending fault system, primarily affecting the plate's northwest and southeast margins, while the eastern Arabian Plate remained relatively undeformed due to the southeastward drift of the Indian Plate. In the Late Cretaceous (~87 Ma), the northwestward movement of the Indian Plate introduced a transtensional regime characterized by NW-SE maximum horizontal stress, resulting in en echelon strike-slip faulting. The onset of Tertiary compression in the mid-Tertiary, driven by continued plate convergence, reoriented the stress field to a NE-SW direction. This tectonic phase led to the formation of the Zagros orogenic belt and widespread low-relief folding across the eastern Arabian Plate. The NE-SW compressional stress persists today, controlling the current regional stress orientation. (Fig. 2).

The study area is situated in the southeastern Arabian Plate, within a region dominated by salt halokinesis and extensively developed salt domes rooted in the Infra-Cambrian basement. Influenced by successive tectonic events—Jurassic rifting, Late Cretaceous transtension, and Tertiary compression—the drape folds above the salt domes display diverse structural styles and host multiple, superimposed fracture sets. The regional conceptual fracture model is defined by dominant NW-SE faulting, accompanied by secondary NE-SW fractures and a complex network of salt-induced fracture systems. (Filbrandt et al., 2006).

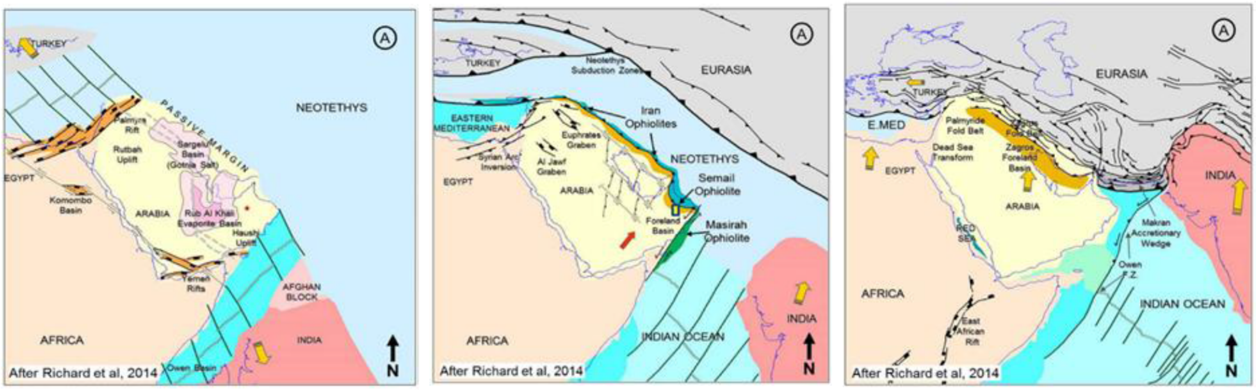

Figure 2—Three major tectonic events in the Arabian Plate: the Jurassic rifting, the Late Cretaceous transtension, and the Tertiary compression (Richard et al., 2017).

To investigate the local structural evolution within the study area, seismic interpretation was conducted to validate the presence and impact of three major tectonic events - Jurassic rifting, Late Cretaceous transtension, and Tertiary compression - on the structural architecture and associated fracturing. A representative seismic profile covering stratigraphic intervals from the Permian to the Tertiary (Pre-Khuff, Arab-D, Hith, Thamama, Mishrif, Simsima, and Rus formations) was analyzed. Thickness variation analysis was employed as a key method to infer tectonic deformation and the timing and development of fault and fracture systems.

The seismic section reveals abundant faulting within the Paleozoic salt dome, while the overlying strata exhibit progressively gentler structural amplitudes upward. The structural crest remains laterally stable across stratigraphic levels, indicating relatively subdued post-Paleozoic tectonic activity. Thickness variation across three key intervals - Top Pre-Khuff to Top Arab-D, Top Hith to Top Mishrif, and Top Mishrif to Top Rus - corresponds to the three tectonic phases. The first two intervals display relatively uniform thickness, whereas the Top Mishrif to Top Rus interval exhibits a slight southward thickening, suggesting increased accommodation space due to Tertiary compression (Fig. 3).

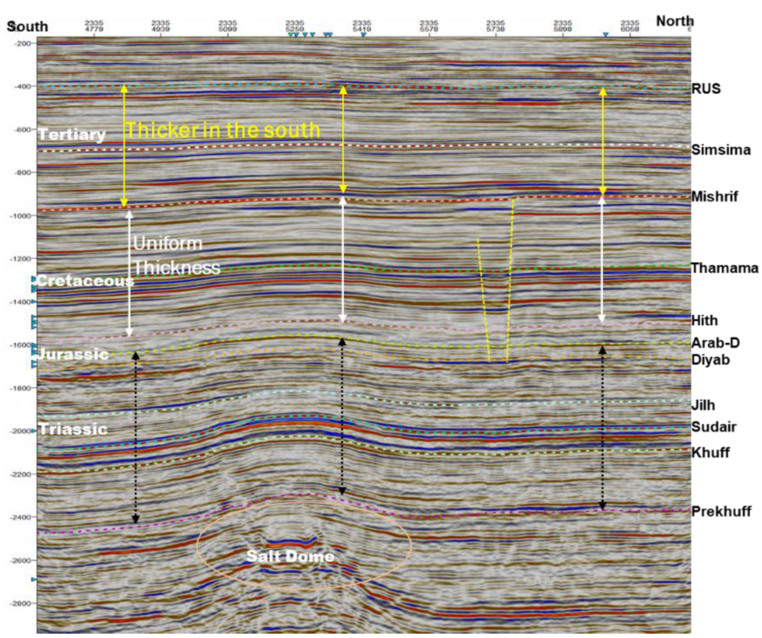

Figure 3—A seismic section with the interpretation from the Permian to the Tertiary shows the structural growth and evolution in the study area.

Detailed isopach mapping further corroborates the tectonic imprints: (1) The Jurassic rifting phase is evidenced by the Hith-Sudair isopach, which shows regional uplift aligned with NW-SE maximum stress, reflecting basement-influenced deformation. (2) During the Late Cretaceous transtension, NW-SE-oriented maximum stress produced contrasting deformation styles. In salt dome regions, the Simsima-Thamama thickness map illustrates elliptical, annular drape folds trending NE-SW due to the obstruction imposed by the salt structures. In contrast, NW-SE en echelon strike-slip faults dominate the non-salt dome areas. These observations suggest that this tectonic phase exerted the primary control on fault development in the study area. (3) The Tertiary compressional regime, which remains active today, resulted in the formation of broad, low-relief anticlines, as observed in the Top Rus structural map. This phase is interpreted as a secondary control on the fracture development, overprinting earlier structural fabrics (Fig. 4).

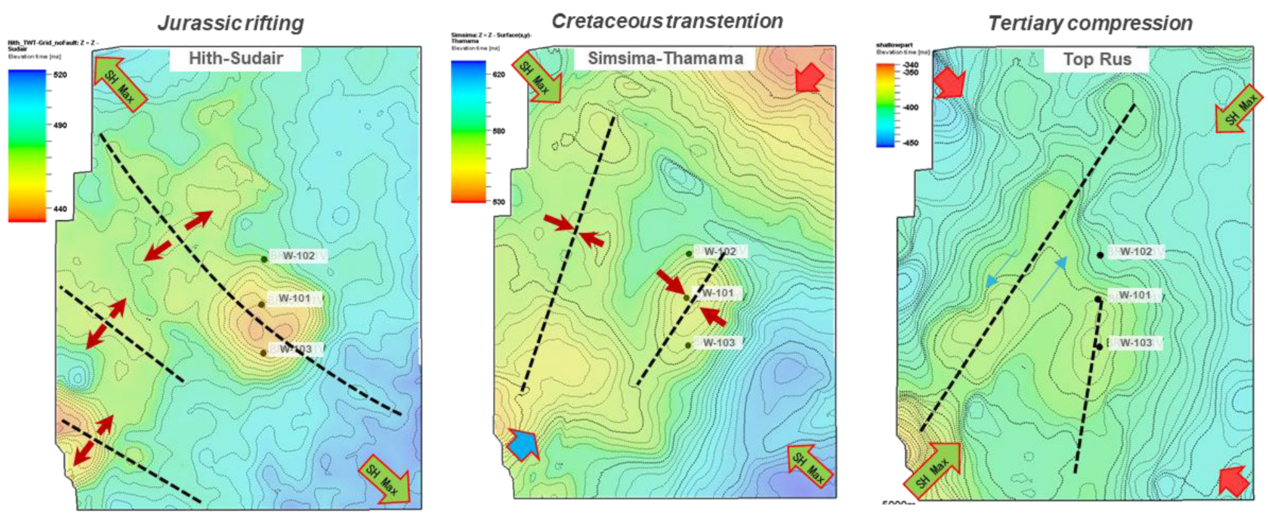

Figure 4—Isopach. Both present the structural growth and evolution in the study area, especially three major tectonic events.

Fracture recognition in salt dome area

Three major tectonic events - Jurassic rifting, Late Cretaceous transtension, and Tertiary compression - have collectively imprinted multiple fracture sets on the drape folds above salt domes, resulting in a structurally complex fracture system in salt-influenced regions. This study proposes a process-oriented workflow to investigate the multi-phase faulting and fracturing history and to characterize the resulting fracture systems in salt dome settings. The methodology begins with an analysis of regional structural evolution to construct a conceptual fracture model reflecting the influence of different tectonic regimes. Subsequently, borehole data - including core observations and borehole imaging logs - are utilized to identify subseismic-scale fractures, including their orientations, dips, and inferred stress regimes, serving to validate and refine the conceptual model. Guided by this model, a suite of seismic attributes is generated to detect local structural trends and characterize medium- to large-scale fracture systems. Finally, the interpreted fractures are cross-validated using dynamic data, including well tests and pressure measurements, and further analyzed within a geological framework to better understand the temporal evolution and overprinting of fracture sets. This integrative approach offers new insights into fracture development in salt dome provinces and enhances prediction capability for fracture-controlled reservoir behavior (Fig. 5).

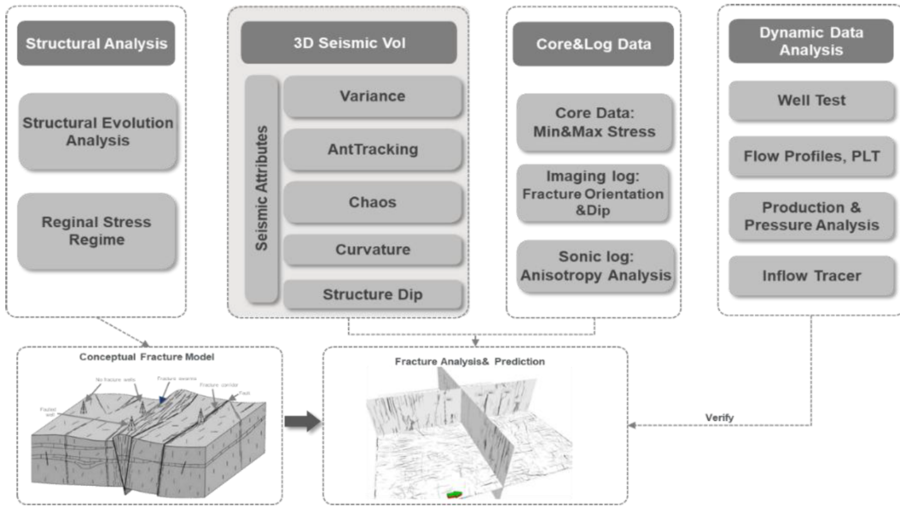

Figure 5—A workflow of integrated fracture analysis and prediction guided by conceptual model through structural evolution analysis.

Core & log analysis

5 wells were drilled at the different locations of the salt dome and 3 wells of W-101, W-102, and W-103 are cored in the study interval (Fig. 4). The core observations show that fractures widely develop in all three wells while dense fractures are observed at the northern (W-102) & southern (W-103) flank, and less fractures are at the dome crest (W-101). The core also shows that the study interval mainly develops high-angle fractures with angles up to 89.6° (Fig. 6).

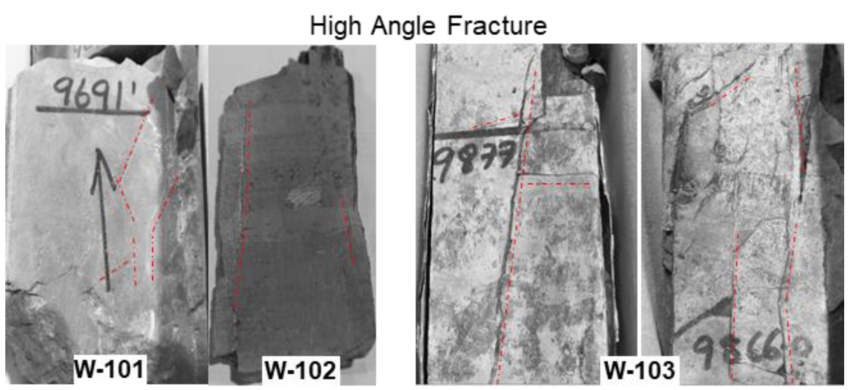

Figure 6—Wide fracture development with high angle up to 89.6° observed on the core of three wells (W-101, W-102 and W-103).

Image logs were collected at the three wells of W-103 (vertical well), W-104 (vertical well), and W-120 (horizontal well) in the study area. The image log data of the three wells acquired in different years and with different approaches, presents variable data quality. The well of W-120 at the crest of the salt dome presents the best quality of the image log. The log interpretation shows that the study interval mainly develops medium-high angle fractures, and the fracture orientation is NW-SE, and the more fractures developed within the subinterval of D3 (Fig. 7). The well of W-104 at the northern flank of the salt dome shows medium data quality. The interpretation shows that the medium-angle fractures developed with the fracture orientation of NW-SE, NNW-SSE, and NE-SW in the study interval while no obvious vertical regularity presents in fracture development. The well of W-103 at the southern flank of the salt dome shows poor data quality. The interpretation shows that the high-angle fractures developed within the study interval, and the fracture orientation is mainly in the NNE-SSW, NE-SW, and NEE-SWW, and the fracture development is irregular vertically (Fig. 7). The interpretation of the induced fractures at W-103 and W-104 indicates that the maximum stress is NE-SW which is consistent with the regional structural evolution analysis. In general, NW-SE fractures are developed at the crest of the salt dome while multiple groups of fractures are developed on both the north and south wings. The complex fault systems result from the multiple phase tectonic deformation. The image logs at the 3 wells show variable data quality might bring uncertainties to the fracture detection.

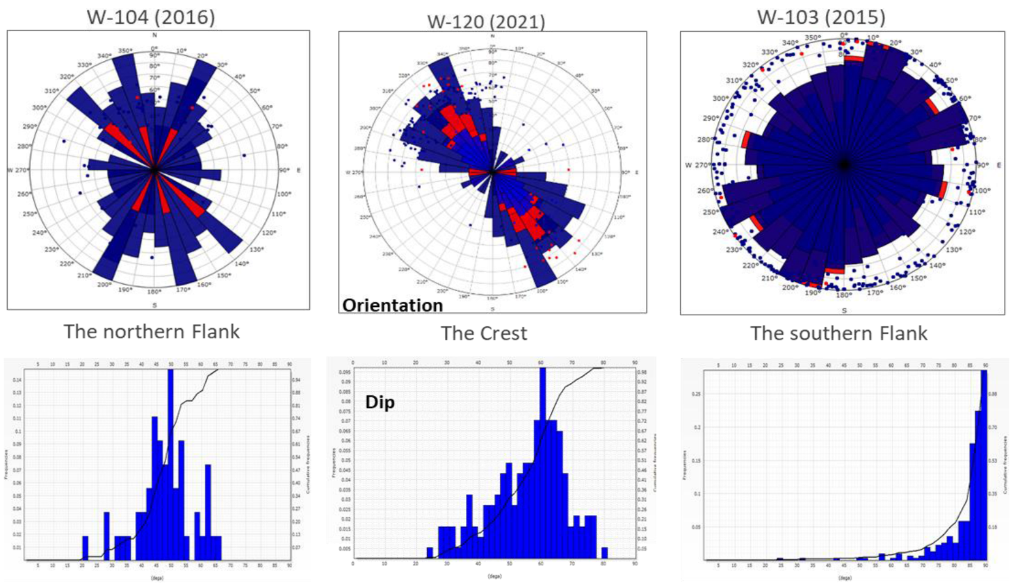

Figure 7—The imaging log and interpretation at the three wells of W-103, W-104 and W-120.

Seismic fracture prediction

The analysis of regional structural evolution indicates that the Cretaceous transtensional phase represents the dominant tectonic event influencing the study area. This tectonic regime primarily generated NW–SE oriented faults and exerted a significant impact on the development of salt dome–related fault systems. Vertically, the structural expression of the transtension is most pronounced within the Lower Cretaceous Thamama Formation, extending downward to the Arab-D Formation and progressively weakening upward toward the Mishrif Formation (Fig. 3). Laterally, the transtensional deformation exhibits spatial heterogeneity: it is most prominent in the northern part of the study area, decreases in intensity toward the south, and intensifies again toward the southwest. Based on the spatial distribution of Cretaceous transtension and the geometry of salt domes, the study area is subdivided into three structural domains. In Regions 1 and 3, NW–SE trending faults dominate, including well-developed en echelon strike-slip faults in the northeastern sector. In contrast, Region 2 is characterized by an elliptical fault pattern associated with the salt dome, formed under the influence of NW–SE maximum horizontal stress, reflecting the interaction between tectonic forces and halokinetic processes (Fig. 8).

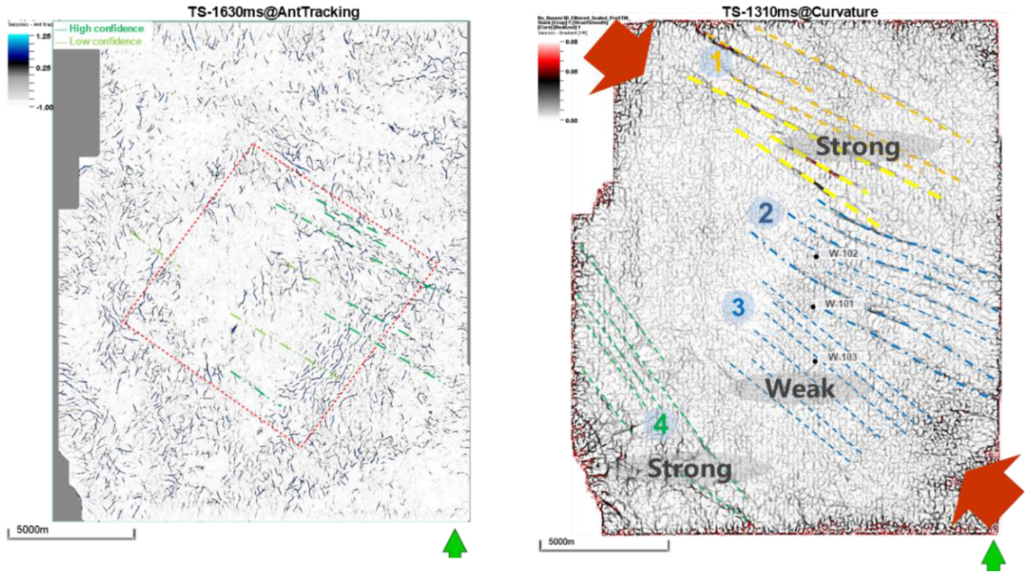

Figure 8—The NW-SE fractures controlled by the Cretaceous transtension within the study interval (Left) and the Cretaceous (Right).

To investigate the fracture system in the study interval, a variety of seismic attributes (variance, curvature, chaos, and ant tracking, etc.) are conducted and combined with seismic sections and structural maps to systematically analyze the evolution of the fault system and the fracture distribution. Under the influence of multi-phase tectonic movement and the salt dome, the three groups of faults and fractures develop in the Upper Jurassic in the study area. 1. NW-SE fracture system. This type of fracture is caused by the major tectonic event of the Cretaceous transtension. In the study area, the mild tectonic movement has less impact on the underlying Upper Jurassic and the NW-SE faults/fractures develop in the study interval, but mostly in the medium and small-scale. The lateral distribution of the NW-SE fault/fracture is consistent with those of regional structural development, with the strongest development in the northeast of the study area and gradually weakening towards the southwest (Fig. 8). 2. Salt dome related fractures. This type of fracture mainly develops in the overlying draped fold along the salt dome. The salt dome related fractures were controlled by the Cretaceous transtension mainly and the Tertiary compression secondly (Fig. 9). Due to the relatively mild tectonic movement in the later period, the salt dome related fractures developed in the deep part and gradually weakened until disappeared in the shallow interval. In the study area, the NW-SE Cretaceous transtension is obviously stronger than the NE-SW Tertiary compression, so the fractures laterally distribute annular or nearly annular along the salt dome, and more strongly at the flanks than at the crest. The small scale of the salt-related fractures develops more intensely in the east & south than in the west & north (Fig. 9). 3. NE-SW fracture system. The NE-SW fault/fracture controlled by the Tertiary compression is a secondary fault in the study area. It mainly develops in the south of the study area and gradually weakens towards the north. Due to the small scale, the NE-SW fracture is barely recognized in seismic section but can be identified by seismic attributes. In the study interval, NE-SW trending faults only develop in the south. Most of the NE-SW fractures are small and medium-scale fractures, which are widely distributed in the study area (Fig. 10).

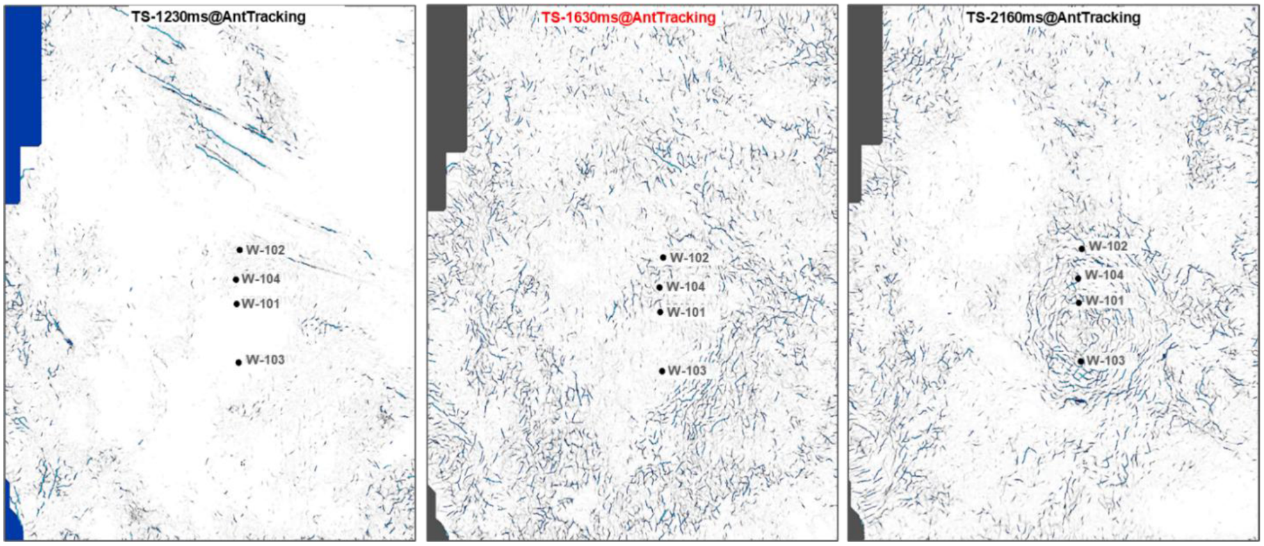

Figure 9—The evolution of the salt dome related fractures within the overlying interval (Left), the study interval (Middle) and the underlying interval.

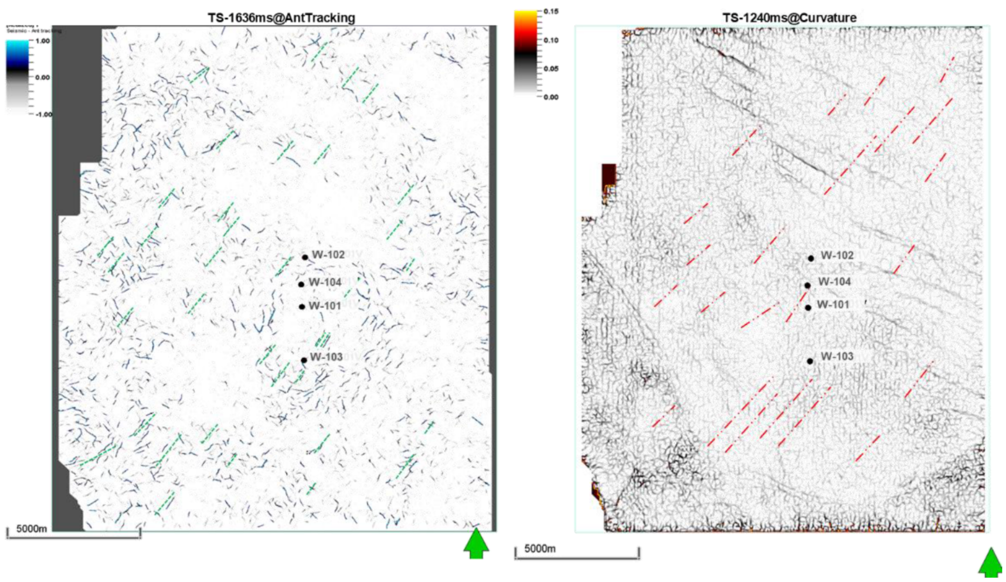

Figure 10—The NE-SW fractures controlled by the Tertiary compression within the study interval (Left) and the Tertiary (Right).

Based on seismic attribute analysis, regional structural evolution analysis and core & image log data, the fracture development characteristics of the Upper Jurassic in the study area are as follows: The main fracture is a NW-SE fracture, which develops in the northeast area of the study area and gradually weakens towards the southwest. The NE-SW fractures are secondary faults, widely developed, but in small scale and difficult to identify. Salt-related fractures are elliptically distributed and mainly developed along the flank, with the most developed fractures in the southeast (Fig. 11).

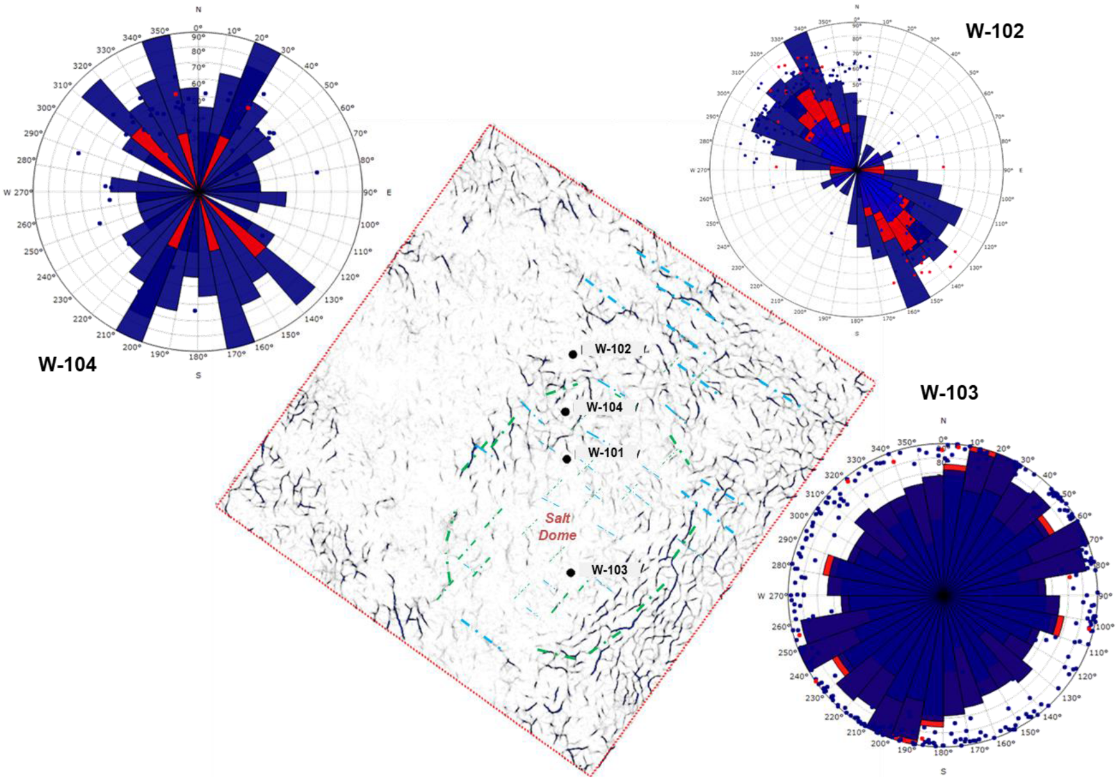

Figure 11—The fracture prediction by the integration of seismic attribute (Ant tracking) and image log. The fracture map shows less fracture develops at the crest of the salt dome while more fractures develop at the flank in the circular geometry. The NW-SE fractures induced by the Late Cretaceous transtension, distribute densely in the northeast of the study area. These recognized fractures generally are consistent with the imaging logs at the different locations relative to the salt dome.

Conclusions

In this study, we conclude that the conceptual fracture model developed through regional and local structural evolution analysis significantly improves the understanding of fracture development in salt dome–dominated areas. The multi-phase tectonic deformation, including Jurassic rifting, Cretaceous transtension, and Tertiary compression, has resulted in a complex and diverse fracture system that exhibits distinct spatial patterns depending on the position relative to the salt dome. Guided by this conceptual model, the integration of seismic attribute analysis and borehole imaging effectively characterizes the fracture system and provides valuable insights into fracture distribution and geometry within the drape folds above the salt domes.

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